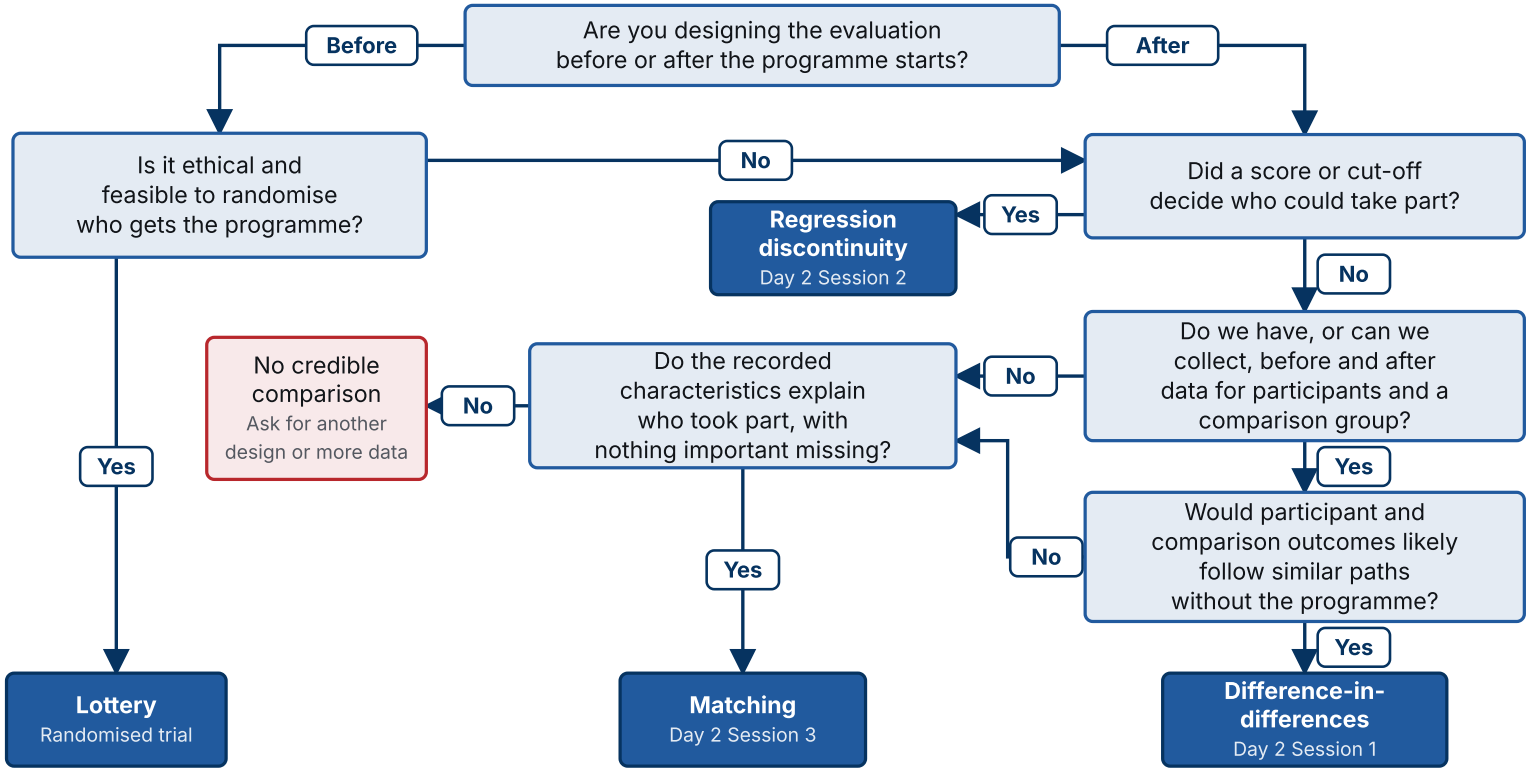
Which method fits? A decision tree for four impact evaluation methods

Start at the top and answer each question for the evaluation you are reading or commissioning. GreenWaste training, Module 2, October 2026.



This tree is a starting guide. It helps you think through what each method needs and what would make it impossible. The order of questions follows how credible each method usually is, but other methods can still be possible: for example, with earlier data and no cut-off you could still plan a difference-in-differences. Every method also needs enough data and a large enough sample, especially when choosing between two options such as regression discontinuity and difference-in-differences. Evaluations are best planned before any data exist, whichever method is chosen.

Lottery (randomised trial)	Groups were assigned by chance. Ask how the draw was done and who dropped out.
Difference-in-differences	Compares the change in participants with the change in a comparison group. Ask whether earlier measurements show similar paths.
Regression discontinuity	Compares participants just either side of the cut-off. It describes those near the cut-off. Ask whether the result holds in a wider window.
Matching	Compares participants with similar non-participants. Ask what was not measured and how close the matches are.
No credible comparison yet	Ask for another design or more data. Until then, read any difference as descriptive, not as the effect of the programme.

What each method needs, and when it can still be planned

Several methods can be possible at once. Use the cards to check the needs and what would rule a method out.

	Lottery (randomised trial)	Regression discontinuity	Difference-in-differences	Matching
What it needs	Assign who gets the programme by chance, before it starts.	A score and cut-off that decided who took part, and enough units near the cut-off.	Outcome measured before and after, for participants and a comparison group on a similar path.	Recorded characteristics that explain who took part, and similar non-participants.
What rules it out	Programme already running, must cover everyone, or the groups cannot be kept apart.	No score or rule, a cut-off that was not followed, or scores that could be gamed.	No comparison group, no way to measure before, or groups already on different paths.	Take-up driven by something unrecorded, such as motivation, or no similar non-participants.
Can still be planned if	Rollout is phased or places are limited, so a draw can decide who goes first.	Eligibility is not set yet, so a score and cut-off can decide who joins.	Earlier data or a cut-off are missing: plan a baseline and follow-up for both groups.	Collect the key characteristics for both groups; combine with the before and after data if they exist.

Several methods can be possible at once. Each also needs enough data and a large enough sample, so planning early keeps more options open.

The tree covers the four methods taught in this module. Other designs exist; the questions point to the assumption each method rests on, not to a guarantee that the method was done well. Print this page (landscape) to save it as a PDF.